MATLAB ile Pipelined ADC'nin Karakterizasyonu
Characterization of Pipelined ADC Using MATLAB

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***Özetçe*— Bu makale 1.5-bit/stage pipeline ADC'lerin performansını geliştirmenin eşik değerleriyle oynayarak mümkün olup olmadığını keşfetmek için yapılan deneylere odaklanmaktadır. Bu makale, 1.5-bit/stage pipelined ADC'sindeki eşik değerlerini değiştirmek ve çıktı üzerindeki etkileri inceleyerek ADC'lerin performansını artırmaya çalıştığımız ADC alanına katkıda bulunmayı amaçlamaktadır. Bunu gereçkleştirmek için hem simülasyon hem de deneyler için MATLAB ile pipelined ADC modeli yapılmıştır.**

Keywords — pipeline ADC, eşik değeri, analog dijital dönüştürücü.

***Abstract*— This paper mainly involves experimentation with threshold values of 1.5-bit/stage pipeline ADCs, to determine if it's possible to make selections for threshold values that improve the overall performance. This paper aimed to contribute to the field of ADCs where we attempted to improve the performance of ADCs by achieving one main objective, which was altering the threshold values in 1.5-bit/stage pipeline ADC and studying the effects on the output. To do that, a pipeline ADC model was built in MATLAB to simulate, test and demonstrate the experiment. This study was entirely realized in MATLAB.**

Keywords — pipeline ADC, threshold, quantization .

I. Introduction

A 1.5-bit/stage pipeline Analog to Digital converters are algorithmic based converters which utilize 3-regions of interest for calculating the result, this is done by correlating each region with a two-digit binary code. Each region is separated into two threshold values, which are used in comparator unit of the ADC stage. In many applications of 1.5-bit/stage pipeline ADC, the threshold value, $|V_{th}|$ is selected as $V_{ref}/4$. This paper provides results and discussion for the experiments done with a 1.5-bit/stage pipeline ADC model built on MATLAB, focusing on the selection of threshold values to improve the performance of the ADC. Experiments were done with the 12-bit, 14-bit and 16-bit resolutions of the pipeline ADC.

Due to real-life non-idealities such as capacitor mismatch and finite opamp gain, the result of the ADC often contains noise and inaccuracies. The MATLAB model designed for this experiment simulated these mismatch errors. Furthermore, parasitic noises like thermal noise were also simulated, by adding white noise to the system.

Ratio of the mismatch error and other factors like threshold values, reference voltage were also parametrizable in the model.

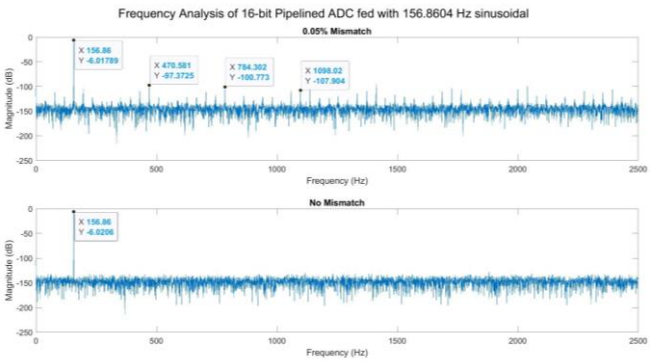
II. 1.5-Bit/Stage Pipeline ADC Model Built With MATLAB

The pipeline ADC model built in MATLAB was used to experiment with threshold values. Being able to select the bit resolution, threshold values and reference voltages of the model was parametrizable. The ADC model basically consists of two main functions, one of them being a sub function of the other, the first function StagePipelinedADC() was designed to calculate a single stage's decision and residue.

The second function is ADCsample() which quantizes a single sample from input signal where it uses the first function to obtain and store the decision codes for each stage, and later performs bitwise operations and normalization to generate the final quantized output.

Additionally, mismatch error can be given as input for ADCsample() function. Since StagePipelinedADC() function inherently uses ADCsample() function, it also takes a vector of mismatch errors as input, which holds the information for the mismatch errors on each stage. This feature provides a possibility to use different mismatch errors for each stage. Figure 1 shows an example of quantized sinusoidal with 16-bit resolution, using the MATLAB model.

Figure 1. Sinusoidal Quantized with MATLAB Model



III. EXPERIMENTAL RESULTS

What was first done in the experiments was to keep the threshold value fixed on each stage, while gradually increasing it from 0.1 V to 0.5 V. This threshold sweep was executed for five different mismatch error ratios:

- 0% mismatch
- 0.020% mismatch
- 0.025% mismatch
- 0.030% mismatch
- 0.035% mismatch

Results for these scenarios are given in Table I, Table II and Table III.

TABLE I. FIXED THRESHOLD SWEEP RESULTS FOR 16-BIT RESOLUTION

16-BIT		**Mismatch Ratio**				
		0	**0.020%**	**0.025%**	**0.030%**	**0.035%**
Highest Performance	**SNDR**	98,1854 dB	91.1322 dB	89.8783 dB	87.9874 dB	87.2831 dB
	|Vth|	0.33 V	0.5 V	0.5 V	0.42 V	0.42 V
Lowest Performance	**SNDR**	96.8101 dB	88.5872 dB	87.2850 dB	84.9197 dB	84.4591 dB
	|Vth|	0.48 V	0.1 V	0.1 V	0.1 V	0.1 V
Improvement	**ENOB**	+0.229 bits	+0.423 bits	+0.431bits	+0.510bits	+0.469bits
	SNDR	+1.375 dB	+2.545dB	+2.593dB	+3.068dB	+2.824dB

TABLE II. FIXED THRESHOLD SWEEP RESULTS FOR 14-BIT RESOLUTION

14-BIT		**Mismatch Ratio**				
		0	**0.020%**	**0.025%**	**0.030%**	**0.035%**
Highest Performance	**SNDR**	86.3068 dB	84.8542 dB	84.3754 dB	83.7776 dB	82.8246 dB
	|Vth|	0.29 V	0.35 V	0.35 V	0.35 V	0.37 V
Lowest Performance	**SNDR**	85.1819 dB	84.1823 dB	83.5005 dB	82.7559 dB	81.4806 dB
	|Vth|	0.48 V	0.48 V	0.1 V	0.1 V	0.1 V
Improvement	**ENOB**	+0.187bits	+0.112bits	+0.145bits	+0.170 bits	+0.223 bits
	SNDR	+1.135 dB	+0.672 dB	+0.875 dB	+1.022dB	+1.344 dB

TABLE III. FIXED THRESHOLD SWEEP RESULTS FOR 12-BIT RESOLUTION

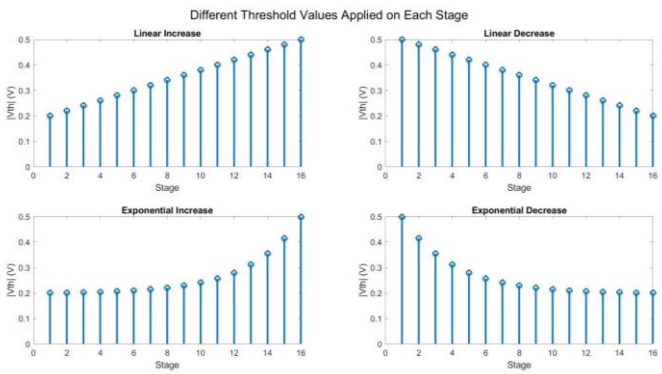
12-BIT		**Mismatch Ratio**				
		0	**0.020%**	**0.025%**	**0.030%**	**0.035%**
Highest Performance	**SNDR**	74.0244 dB	73.7670 dB	73.7173 dB	73.6593 dB	73.5704 dB
	|Vth|	0.1 V	0.12 V	0.27 V	0.29 V	0.31 V
Lowest Performance	**SNDR**	72.9620 dB	72.8086 dB	72.8604 dB	72.8139 dB	72.7562 dB
	|Vth|	0.46 V	0.46 V	0.46 V	0.46 V	0.46 V
Improvement	**ENOB**	+0.177bits	+0.159bits	+0.142bits	+0.140bits	+0.135bits
	SNDR	+1.062dB	+0.958dB	+0.857 dB	+0.845dB	+0.814dB

When it was done with fixed thresholds, the idea of using different thresholds in each stage of the ADC was tested. For that, a certain range for thresholds was determined, which was between 0.2 V to 0.5 V. Then, by applying four different schematics that follow this range was implemented on the MATLAB model. As mentioned above, there were four different schematics:

1. Linearly Increasing
2. Linearly Decreasing
3. Exponentially Increasing
4. Exponentially Decreasing

Schematics are shown in Figure 2.

Figure 2. Four Threshold Schematics Applied to Stages of Pipeline ADC



Overall results, including fixed threshold scenarios, are presented in Table IV.

TABLE IV. OVERALL RESULTS OF THRESHOLD EXPERIMENTS

OVERALL RESULTS		Mismatch				
		0	**%0.020**	**%0.025**	**%0.030**	**%0.035**
16-bit SNDR	**Linear Increase**	96.4977 dB	89.4988 dB	88.1994 dB	86.1351 dB	85.5349 dB
	Linear Decrease	98.1064 dB	91.1876 dB	89.8412 dB	87.9894 dB	87.3284 dB
	Exponential Increase	97.2164 dB	89.6254 dB	88.2608 dB	86.1154 dB	85.5047 dB
	Exponential Decrease	98.1064 dB	91.1938 dB	89.8235 dB	87.9675 dB	87.2934 dB
	Fixed Threshold	98,1854 dB	91.1322 dB	89.8783 dB	87.9874 dB	87.2831 dB
14-bit SNDR	**Linear Increase**	84.7196 dB	83.3275 dB	82.8820 dB	82.3354 dB	81.4557 dB
	Linear Decrease	86.2703 dB	84.9846 dB	84.5186 dB	83.9881 dB	82.9952 dB
	Exponential Increase	85.5520 dB	84.0250 dB	83.4886 dB	82.8541 dB	81.8635 dB
	Exponential Decrease	86.2703 dB	84.9823 dB	84.5010 dB	83.9545 dB	82.9998 dB
	Fixed Threshold	86.3068 dB	84.8542 dB	84.3754 dB	83.7776 dB	82.8246 dB
12-bit SNDR	**Linear Increase**	72.3083 dB	72.2303 dB	72.2569 dB	72.2312 dB	72.1417 dB
	Linear Decrease	73.9229 dB	73.7712 dB	73.7611 dB	73.7073 dB	73.6300 dB
	Exponential Increase	72.9879 dB	72.9271 dB	72.9546 dB	72.9132 dB	72.8131 dB
	Exponential Decrease	73.9229 dB	73.7723 dB	73.7597 dB	73.7073 dB	73.6289 dB
	Fixed Threshold	74.0244 dB	73.7670 dB	73.7173 dB	73.6593 dB	73.5704 dB

IV. DISCUSSION

A. *Fixed Threshold Sweep*

In Table I, pipeline ADC with 16-bit resolution was simulated with five different mismatch ratios. According to the same table, it's possible to see there were some sweet spots for threshold selection detected. Although we can say this sweet spot may vary according to the mismatch error's ratio, this sweet spot was mostly observed between 0.4 V – 0.5 V. However, in an ideal case, where there is no mismatch error, best selection for the threshold seems to be around 0.33 V. Last two rows of the table present the difference of efficiency, between the worst pick and the best selection of threshold. It's possible to say whenever mismatch noise was introduced to the system, potential to improve the performance by selecting the right threshold value increased. However, the results also indicate that improvement in ENOB was mostly around 0.5 V, which cannot be countered as an exceptional outcome.

Results for 12-bit and 14-bit resolutions can be seen in Table II and Table III. For 14-bit scenarios, it was observed that 0.35 V seemed to be a better choice for $|V_{th}|$, while 0.48 V and 0.1 V giving the least efficient performance. In 12-bit resolutions, the sweet spot for the threshold selection seemed to be smaller, compared to other scenarios which utilized different bit resolutions. However, for both 14-bit and 12-bit ADC, altering the threshold value gave similar performance improvements, which was not exceptional.

B. *Applying Different Threshold Values on Each Stage*

Figure 2 shows the four different schematics used to determine threshold values for each stage. In Table IV, SNDR for all scenarios were presented. Results show that while it's not easy to tell whether exponential decrease or linear decrease is a better choice, decreasing schematics in general seemed to give the best performance, among four different schematics (Excluding fixed threshold scenarios). However, like it happened with fixed threshold sweep experiments, differences in SNDRs for different scenarios are not remarkable.

C. *Overall Results*

Table IV also includes the best selection for fixed threshold scenarios. In many cases, fixed threshold value seemed to give better performance, however, it was not very different from using different thresholds on each stage, in the context of performance.

V. CONCLUSION

1.5-bit/stage pipeline ADC algorithm is easy and suitable for simulation, thus making this experiment easy to execute as well. If these experiments were made with real hardware, results might have been different with small margins in mismatch error. However, this wouldn't dramatically affect the results of the experiment, thus, this paper can be insightful to have an understanding of how selection threshold values with different patterns might affect the outcome.

For most cases, fixed threshold value seemed to be a better option, while for some cases applying different thresholds to each stage gave the better performance. However, in real life implementations, the cost of utilizing such schematics may not be worth the improvement in the performance, as it was very small. Furthermore, in some cases, mismatch errors affected the performance of different schematics, meaning knowledge of the mismatch error's value would be also important, if it was decided to implement such schematics. And if the mismatch error was known, calibrating the ADC would be a better choice in general.

ACKNOWLEDGMENT

We would like to thank Istanbul Medipol University for providing us with MATLAB licenses. And we also give appreciation to our advisor, Dr. Mustafa Aktan for his guidance throughout our project.

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